# Finite terms: check, DGLAP and running coupling, simplification, final result, and comparison with arXiv:1411.2869

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## 1 Summary

Your expressions are in black, removed text in red, new or changed text in blue. "The file" is `Finite_terms.tex` as received; "the paper" is T. Altinoluk, N. Armesto, G. Beuf, A. Kovner, M. Lublinsky, *Single-inclusive particle production in proton-nucleus collisions at next-to-leading order in the hybrid formalism*, arXiv:1411.2869 (AABKL).

**1. Are these the final terms?** Yes, with the corrections F1–F7 of Sec. 3. Every row was checked against the exact $p^+$ integral of its (earlier verified) first-principles integrand, with no limit in $\vee$, $k^+$ or $\Lambda$. The test is that "exact minus file" must be (coefficient)$\times \log \frac{\vee}{\Lambda}$ for every $\vee$ and $\Lambda$, i.e. the file contains everything except the evolution term. This holds to $10^{-17}$–$10^{-12}$ for the rows with closed forms, and to $O(\Lambda/k^+)$ for the rows whose plus prescriptions start at 0 instead of $\Lambda$ (Table 1).

**2. What "finite" contains.** With your convention ($\log \frac{\Lambda}{k^+} = -\log \frac{\vee}{\Lambda} + \log \frac{\vee}{k^+}$, rule A), the file still contains large logarithms $L_\vee \equiv \log \frac{\vee}{k^+}$: explicitly, and hidden in $\log \frac{(\vee - k^+)A + k^+ B}{\Lambda A + k^+ B}$, in the plus-prescribed $p^+$ integrals up to $\vee - k^+$ and in the $\xi$ integral of Group II Row IV that starts at $k^+/\vee$. Their total coefficient is $\mathcal{N}(\mathcal{B} - \mathcal{J})$ (BFKL⊗LO minus JIMWLK⊗LO; `LogVeeKplus_BFKL.pdf`). Row by row it equals (region-P limit − region-T limit) of the integrand; this was re-checked here (Sec. 7). At mid rapidity $L_\vee = \frac{1}{2} \log \frac{\vee}{\Lambda}$, so this piece is *not* small (question Q1).

**3. Segregation.**

- **Running coupling** (Sec. 5): Group I Row II part 1 ($\beta_0$, and the constant $N_c(\frac{67}{9}-\frac{\pi^2}{3})$) and Group I Row V part 2b UV ($\beta_0$). The $\beta_0$ logarithms cancel between them ($+2-2$ in units of $\frac{\alpha_s\beta_0}{8\pi}(\ell_X+\ell_{X'})$), so the scale of $\alpha_s$ in LO is fixed by the fragmentation scale. What is left is $\frac{\alpha_s}{2\pi}N_c(\frac{67}{9}-\frac{\pi^2}{3})\,\mathrm{LO}$.
- **DGLAP** (Sec. 6): the first two lines of Group II Row IV, plus the $\delta(1-\xi)$ pieces of Group I Rows II and V and the charge renormalization of LO. The pole is $\frac{\alpha_s}{2\pi}(-\frac{2}{\epsilon})P_{gg}\otimes\mathrm{LO}$ and is absorbed in $D_{h/g}$. The finite rest is $-\frac{\alpha_s}{2\pi}P^{(r)}_{gg,+}\otimes\ell_\Delta\mathrm{LO}$.

**4. Simplification** (Secs. 8, 9).

- $\Lambda\to 0$ and $\vee\to\infty$ with $L_\vee$ separated. Every closed-form row then reduces to logarithms of transverse distances; the error is $O(k^+/\vee,\Lambda/k^+)$, checked at $\vee/k^+=10^6,10^9$.
- For the Group III rows with the splitting after the target (Row III parts 1 and 2, Rows V, VI), the large-$p^+$ region sits at $y'\to z$ in the variables of the file, where $\mathcal{K}(y'-z)$ is singular; $\vee\to\infty$ must not be taken pointwise in $y'$ there. In the variables $w'$ both limits are explicit, and each row is $\frac{1}{2\pi}[\mathcal{T}\log\frac{\vee}{\Lambda}+(\mathcal{P}-\mathcal{T})L_\vee]+R$ with $\mathcal{T}$, $\mathcal{P}$ given in closed form.
- In Group II Row I part 2, the parts 2a, 2d, 2e, 2f carry no $L_\vee$.
- **Exact cancellation:** Group I Row IV $=-$Group III Row I, for every $p^+$ and every configuration (symmetrized charges). **Combination:** Group II Row I part 1 + Group III Row II collapse to one term with colour $\mathcal{C}^a_L\rho^b(y)\mathcal{R}^{ab}(w,z,x)$.
- **The remaining symmetrized three-$\rho$ finite terms do not cancel** (e.g. the $U(z)$ network differs by $O(1)$, since the energy denominators are $p^+A-k^+B$ in the virtual and $p^+A+k^+B$ in the real rows). Their $L_\vee$ parts do cancel. They are the classical first saturation correction of the projectile; they vanish after a Gaussian (MV) average over the projectile (question Q2). Their commutator parts are two-$\rho$ terms and must be kept.

**5. Final result:** Eq. (10.1).

**6. Comparison with AABKL** (Sec. 11). Your result has the structure of their gluon channel, Eq. (4.22) of the paper, with the incoming collinear gluon replaced by the gluon that the projectile charges emit:

- the same $g\to gg$ vertex and $P_{gg}$;
- the same three splitting topologies: before the target ($s_A[y,\bar y]$), after the target ($(1-\xi)^2 s_A[(1-\xi)y,(1-\xi)\bar y]$), and their interference ($sss$ and $X$ terms);
- the same virtual dressed-gluon term, Eq. (6.17) of the paper;
- the fragmentation counterterm.

The differences:

- a linearly polarized parent gluon (the $d_\perp$ term);
- the emission of the parent after the target, by the rotated charge;
- $\beta_0$ terms;
- no PDF-type collinear subtraction: it is replaced by the emission distance $|x-w|$;
- a cutoff on $p^+$ instead of the Ioffe-time cutoff on $(\xi,l_\perp)$.

**7. Questions:** Sec. 12.

# 2 Notation

- $L_\vee\equiv\log\frac{\vee}{k^+}$; $\ell_r\equiv\log\frac{r^2\mu^2_{\overline{MS}}}{c_0^2}=2\gamma_E+\log\frac{r^2\mu^2_{\overline{MS}}}{4}$, $c_0=2e^{-\gamma_E}$; $\hat E_r\equiv\frac{2}{\epsilon}+\ell_r$ (the bracket $\frac{2}{\epsilon}+2\gamma_E+\log\frac{r^2\mu^2}{4}$ of the file).
- $A\equiv(x-z)^2$, $B\equiv(x-w)^2$, $A'\equiv(x'-z)^2$, $B'\equiv(x'-w')^2$, $D\equiv B'A-BA'$.
- $X'\equiv\frac{x'-w'}{(x'-w')^2}$, $Y\equiv\frac{y-z}{(y-z)^2}$, $X_w\equiv\frac{x-w}{(x-w)^2}$, $X_z\equiv\frac{x-z}{(x-z)^2}$, $Z_w\equiv\frac{z-w}{(z-w)^2}$.

- Colour: $\mathcal{C}_L^a \equiv U^{ab'}(x')\rho^{b'}(x') - U^{ab'}(w')\rho^{b'}(x')$ and $\mathcal{R}^{ab}(w,z,x) \equiv f^{cbd}U^{ac}(w)\rho^d(x) - f^{acd}U^{bc}(z)U^{de}(x)\rho^e(x)$.
- The bracket of Group I Row IV, Group II Row I and Group III Rows I, II:

$$\mathbb{T}^{ij} \equiv \frac{\delta^{ij}}{2(z-w)^2} + \frac{(x-z)^j(z-w)^i}{(z-w)^2(x-z)^2} - \frac{(x-z)^j(x-w)^i}{(x-w)^2(x-z)^2} - \frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2}.$$

- LO and its fragmentation form:

$$\begin{aligned}\mathrm{LO}\big[f\big] \equiv \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2 k^+}\int_{x,x',w,w'} e^{-ik\cdot(w'-w)}\, f\, \frac{(x-w)\cdot(x'-w')}{(x-w)^2(x'-w')^2} \\ \times \big[U(w')-U(x')\big]^{ab'}\rho^{b'}(x')\big[U(w)-U(x)\big]^{ab}\rho^b(x),\end{aligned}$$

$$\mathrm{LO}_\xi\big[f\big] \equiv \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2 k^+}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\, f\, \frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\,\mathcal{O},$$

  with $\mathcal{O} = [U^{e'c}(y') - U^{e'c}(x')][U^{ec}(y) - U^{ce}(x)]\rho^{e'}(x')\rho^e(x)$ and $\mathrm{LO} \equiv \mathrm{LO}[1]$.
- $\mathcal{N} \equiv \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}$; $\mathcal{J}$ = JIMWLK⊗LO, $\mathcal{B}$ = BFKL⊗LO in units of $\mathcal{N}$, as in `Final_evolution_check.pdf`.
- $U^{bc} = U^{cb}$, as in your files; charges at different points commute.

# 3 Check of the file

## 3.1 Method

For every row the first-principles integrand (earlier notes: `fp1.py`, `fixed23.py`, with the corrected signs E3 and M1) was integrated exactly over $p^+$, with principal values where needed, on its exact range: $\Lambda < p^+ < \vee - k^+$; $\Lambda < p^+ < k^+ - \Lambda$ and $k^+ + \Lambda < p^+ < \vee$ for Group I Row III; $\Lambda < p^+ < k^+ - \Lambda$ for Group I Row V. The transverse points, charges and SU(3) Wilson lines are random; the difference

$$D(\vee, \Lambda) = \text{exact} - \text{file}$$

was computed for $(\vee, \Lambda) = (6, 10^{-3}), (40, 10^{-3}), (6, 10^{-5}), (40, 10^{-5}), (300, 10^{-4})$, $k^+ = 1$. If the file is complete, $D/\log\frac{\vee}{\Lambda}$ is the same number in all five cases.

Table 1: Check of `Finite_terms.tex`. "Spread" is the largest change of $D/\log\frac{\vee}{\Lambda}$ over the five $(\vee, \Lambda)$, relative to the row (three configurations).

| row | compared with | result |
|---|---|---|
| G1 Row II part 1 | corrected Group I file (`Group_I_v2_corrected.pdf`) | identical after removing $-2\log\frac{\vee}{\Lambda}\hat{E}$ |
| G1 Row II 2a, 2b | corrected Group I file | identical (2a: sign of the $f^{abc}$ line correct, remark R); typo F3 |
| G1 Row III ($N_1, N_2, N_{3a}, N_{3b}$) | exact principal-value $p^+$ integral | spread $7\times10^{-13}$ (with $\log\lvert\cdot\rvert$, F1) |
| G1 Row IV | exact $p^+$ integral | spread $2\times10^{-17}$ |
| G1 Row V part 1 | exact on $(\Lambda, k^+ - \Lambda)$ | spread $9\times10^{-5} = O(\Lambda/k^+)$ (F2) |
| G1 Row V part 2 (2a, 2b UV, 2b finite) | `Group_I.tex` (verified in `Group_I_v2_check.pdf`) | identical after removing $\log\frac{\vee}{\Lambda}$; $\Lambda\to0$ in 2b UV |
| G2 Row I part 1 | exact $p^+$ integral | spread $4\times10^{-17}$ |
| G2 Row I part 2 (2a–2f) | exact $p^+$ integral | spread $2\times10^{-17}$ |
| G2 Row IV | `RowIV_FF_userform.pdf` (rel. $2\times10^{-14}$) | identical after removing $\log\frac{\vee}{\Lambda}$; F7 |
| G3 Rows I, II | exact $p^+$ integral | spread $3\times10^{-17}$, $3\times10^{-17}$ |
| G3 Row III part 1, Rows V, VI | exact $p^+$ integral | spread $1\times10^{-4} = O(\Lambda/k^+)$ (F2) |
| G3 Row III part 2 | exact $p^+$ integral | spread $3\times10^{-5} = O(\Lambda/k^+)$ (F2); typo F4 |

The plus-prescription rows were also checked at $\Lambda = 10^{-3}, 10^{-4}, 10^{-5}$: the spread falls with $\Lambda$, as an $O(\Lambda/k^+)$ term must.

## 3.2 Corrections

**F1 Group I Row III: the logarithms are principal values.** The $p^+$ integrand has a pole at $p^+ = k^+B/A$, so every logarithm with $k^+B - cA$ in it must be read as $\log|\cdot|$. For example, in $N_1$:

$$\log\left[\frac{\vee A - k^+B}{(\Lambda+k^+)A - k^+B}\right] \;\to\; \log\left|\frac{\vee A - k^+B}{(\Lambda+k^+)A - k^+B}\right|,$$

and likewise the logarithms of $N_1$ (fourth and fifth terms), $N_2$ and $N_{3a}$. With $\log|\cdot|$ the file agrees with the exact integral to $10^{-12}$.

**F2 Plus prescriptions from** 0 (Group I Row V parts 1 and 2a; Group III Row III parts 1 and 2, Rows V, VI). They are exact only with lower limit $\Lambda$; from 0 they are off by $O(\Lambda/k^+)$. This is harmless for $\Lambda \to 0$, but say so in the text.

**F3 Group I Row II, $d^3N_{2a}$:** the integrand does not depend on $z$: $\int_{x,y,x',z} \to \int_{x,y,x'}$.

**F4 Group III Row III, second part:** $w'$ is integrated out by the $\delta$ function, $\rho(x')$ is in the colour factor and $y$ does not occur. So $\int_{w,w',y',z,y,x} \to \int_{w,x',y',z,x}$ in the first term, and $\int_{w,y',z,y,x} \to \int_{w,x',y',z,x}$ in the $\log(1-\frac{k^+}{\vee})$ term.

**F5** Group I Row II part 1: $\int_{w,w',x,x',}$.

**F6 The four-$\rho$ rows are not exactly "NA".** Group II Rows II, III and Group III Row IV run over $\Lambda < p^+ < \vee - k^+$ with integrand $\propto \frac{1}{p^+}$, so they give $\log\frac{\vee}{\Lambda} + \log(1-\frac{k^+}{\vee})$. The second term times the four-$\rho$ structure is in the finite part; Group I Row I (range $(\Lambda, \vee)$) has none. This is $O(k^+/\vee)$ and can be dropped together with the other power corrections (Sec. 4).

**F7 Group II Row IV, third line:** no pole multiplies it, so $d_\perp = 2$: $\Big[\frac{(x-y)\cdot(x'-y')}{\dots} - d_\perp 2\,\frac{(x-y)\cdot(y'-y)(x'-y')\cdot(y'-y)}{\dots}\Big]$.

**R Remark (not in this file):** in `Group_I.tex`, Row II 2a, the $f^{abc}U^{bd}(x)U^{ce}(y)$ line is written as $+\frac{ig^4 f^{abc}}{32\pi^4}[\log\frac{\vee}{\Lambda} - \log\frac{\vee}{k^+}]$. The correct sign is minus (`Group_I_v2_corrected.pdf`). The finite file has the correct finite part, $+\frac{ig^4 f^{abc}}{32\pi^4}\log\frac{\vee}{k^+}$. The evolution list built from it was checked separately and is correct.

# 4 Power corrections

The following vanish for $\Lambda/k^+ \to 0$, $k^+/\vee \to 0$; they are dropped from here on:

- $\log(1-\frac{k^+}{\vee})$, $\log^2(1-\frac{k^+}{\vee})$, $\log(1-\frac{k^+}{\vee})L_\vee$ (Group I Rows II, III, IV, V; Groups II, III), and the four-$\rho$ rest F6;
- $\log(1\pm\frac{\Lambda}{k^+})$ (Group I Row V 2b UV; the shifts $\vee/(k^+\pm\Lambda) \to \vee/k^+$);
- $\frac{1}{\vee} - \frac{1}{k^+ + \Lambda} \to -\frac{1}{k^+}$ (Group II Row I 2f);
- the lower limit 0 instead of $\Lambda$ of the plus prescriptions (F2);
- the $\frac{k^+}{\vee}$ inside $R^i$ (Group I Row II 2b): $R^i \to R^i_\infty = -2k^i \log\frac{p^2}{k^2} + \frac{(2p^2-2k\cdot p+k^2)k^i + k^2 p^i}{(k-p)^2}\log\frac{k^2}{(k-p)^2}$;
- the remainder $-N_c\log(1-\frac{k^+}{\vee})$ of the $\delta(1-\xi)$ tally (Sec. 6).

Pointwise in the transverse coordinates the logarithms with $\vee$ and $\Lambda$ inside are, exactly up to such terms,

$$\log\frac{(\vee-k^+)A+k^+B}{\Lambda A+k^+B} \to L_\vee + \log\frac{A}{B}, \qquad \log\Big|\frac{\vee A-k^+B}{(\Lambda+k^+)A-k^+B}\Big| \to L_\vee + \log\Big|\frac{A}{A-B}\Big|,$$

$$\log\Big|\frac{(k^+-\Lambda)A-k^+B}{\Lambda A-k^+B}\Big| \to \log\Big|\frac{A-B}{B}\Big|, \qquad \log\frac{[k^+(x-w)-(k^+-\Lambda)(x-z)]^2}{[k^+(x-w)-\Lambda(x-z)]^2} \to \log\frac{(z-w)^2}{B},$$

$$\log\frac{k^+-\Lambda}{\vee} \to -L_\vee, \qquad \mathcal{A} \to \mathcal{A}_0,$$

with $\mathcal{A}_0 = \frac{1}{r}\big[\arctan\frac{A-(x-w)\cdot(x-z)}{r} + \arctan\frac{(x-w)\cdot(x-z)}{r}\big]$, $r = \sqrt{AB-[(x-w)\cdot(x-z)]^2}$.

# 5 Running coupling

**The terms.** In units of LO (inside the integrals), with the c.c. of Group I:

$$\text{G1 Row II, part 1} = \frac{\alpha_s}{2\pi}\,\text{LO}\Big[\beta_0\,\tfrac{1}{2}\big(\hat{E}_X+\hat{E}_{X'}\big) + N_c\Big(\frac{67}{9}-\frac{\pi^2}{3}\Big) + N_c L_\vee\big(\hat{E}_X+\hat{E}_{X'}\big)\Big], \tag{5.1}$$

$$\text{G1 Row V, 2b UV} = \frac{\alpha_s}{2\pi}\,\text{LO}\Big[-\beta_0\,\tfrac{1}{2}\big(\hat{E}_X+\hat{E}_{X'}\big) - 2N_c L_\vee\big(\hat{E}_X+\hat{E}_{X'}\big)\Big] \quad (y\to x,\ z\to w), \tag{5.2}$$

with $X = x-w$, $X' = x'-w'$, $\beta_0 = \frac{11}{3}N_c$. In 2b UV, $[U(x')-U(w')][U(y)-U(z)]$ with $y\to x$, $z\to w$ is the LO colour factor, and $[\frac{2}{\epsilon}+\dots] = +\hat{E}$. The prefactors were checked with `sympy`: the $\beta_0$ parts are $+2$ and $-2$ in units of $\frac{\alpha_s\beta_0}{4\pi}\hat{E}$ (`checks.json`).

**Result.**

- **The $\beta_0$ terms cancel** (pole, $\gamma_E$, $\mu$ and $\log X^2$ together): Row II is $K_{JSJ}$ of KLSZ (arXiv:2308.15545), Row V 2b is the dressing of the measured gluon on $U(w)$ and $U(w')$ (`Final_evolution_check.pdf`, Sec. 9).
- **Left over:**

$$\boxed{\left.\frac{d^3N}{d^3k}\right|_{\rm RC} = \frac{\alpha_s}{2\pi}\,N_c\Big(\frac{67}{9}-\frac{\pi^2}{3}\Big)\,\mathrm{LO}} \tag{5.3}$$

  This is the constant of $K_{JSJ}$ (CMW constant $K = N_c(\frac{67}{18}-\frac{\pi^2}{6})$, times $\frac{\alpha_s}{\pi}$). It is not a logarithm.

- The $L_\vee\hat{E}$ terms of (5.1), (5.2) are not running coupling: with the $L_\vee$ term of Group II Row IV their poles cancel, and the rest is part of $-\mathcal{J}L_\vee$ (Sec. 6).
- **The scale of $\alpha_s$ in LO.** The coupling of LO is renormalized in $\overline{MS}$. Its counterterm is the $\frac{\beta_0}{2}\delta(1-\xi)$ part of the fragmentation counterterm, so the natural scale is the fragmentation scale, $\alpha_s(\mu_F)$ with $\mu_F \simeq c_0/|y'-y| \sim |\mathbf{k}|$. Equivalently (KLSZ's $\lambda=\frac{1}{2}$), to this order

$$\mathrm{LO}\big|_{\alpha_s(\mu_F)} = \mathrm{LO}\Big[\alpha_s \to \tfrac{1}{4\pi}g\big(\tfrac{c_0^2}{X^2}\big)g\big(\tfrac{c_0^2}{X'^2}\big)\Big] - \frac{\alpha_s\beta_0}{8\pi}\,\mathrm{LO}\big[\ell_X+\ell_{X'}\big]\Big|_{\mu=\mu_F} + O(\alpha_s^3).$$

# 6 DGLAP

**Group II Row IV, first two lines.** With $d_\perp = 2-\epsilon$ your weight is $W = -\frac{2}{\epsilon} - \ell_\Delta + O(\epsilon)$, $\Delta = y'-y$. Your subtraction at $\xi=1$ converts exactly to the standard plus distribution (`RowIV_FF_userform.pdf`, Eq. (3.3)), so lines 1 and 2 of the file are

$$\frac{\alpha_s}{2\pi}\int_{\frac{k^+}{\vee}}^1 \frac{d\xi}{\xi^2}\,P^{(r)}_{gg,+}(\xi)\,\mathrm{LO}_\xi\big[W\big] \;-\; \frac{\alpha_s}{2\pi}\,2N_c\,L_\vee\,\mathrm{LO}_1\big[W\big], \qquad P^{(r)}_{gg,+} = 2N_c\Big[\frac{\xi}{(1-\xi)_+}+\frac{1-\xi}{\xi}+\xi(1-\xi)\Big]. \tag{6.1}$$

**The poles.** In units of $\frac{\alpha_s}{2\pi}(-\frac{2}{\epsilon})N_c\,\mathrm{LO}_1$, the $\delta(1-\xi)$ terms (unchanged from `Final_evolution_check.pdf`, now without $\log\frac{\vee}{\Lambda}$) are:

| source | coefficient |
|---|---|
| G2 Row IV, Eq. (6.1) | $-2L_\vee$ |
| G1 Row II part 1 (with c.c.) | $-\frac{11}{3}-2L_\vee-\log(1-\frac{k^+}{\vee})$ |
| G1 Row V 2b UV (with c.c.) | $\frac{11}{3}+4L_\vee$ |
| charge renormalization of LO | $+\frac{11}{6}$ |
| sum | $\frac{11}{6}+O(k^+/\vee)$ |

So the full pole is $\frac{\alpha_s}{2\pi}(-\frac{2}{\epsilon})\int\frac{d\xi}{\xi^2}P_{gg}(\xi)\,\mathrm{LO}_\xi$, with

$$P_{gg}(\xi) = 2N_c\Big[\frac{\xi}{(1-\xi)_+}+\frac{1-\xi}{\xi}+\xi(1-\xi)\Big]+\frac{11N_c}{6}\,\delta(1-\xi) \qquad (n_f=0).$$

It is absorbed in $D_{h/g}$: $D^{(0)}_{h/g}(\xi) = \int_\xi^1\frac{dz}{z}\big[\delta(1-z)+\frac{\alpha_s}{2\pi}\frac{2}{\epsilon}P_{gg}(z)\big]D_{h/g}(\frac{\xi}{z},\mu_F)$, and $\mu_F$ independence gives

$$\mu_F^2\frac{\partial}{\partial\mu_F^2}D_{h/g}(\xi,\mu_F) = \frac{\alpha_s(\mu_F)}{2\pi}\int_\xi^1\frac{dz}{z}P_{gg}(z)\,D_{h/g}\Big(\frac{\xi}{z},\mu_F\Big). \tag{6.2}$$

**The $L_\vee$ terms with $\hat{E}$.** Group II Row IV, Eq. (6.1), $+\frac{\alpha_sN_c}{\pi}L_\vee\,\mathrm{LO}[\hat{E}_\Delta]$ (at $\xi=1$, $y\to w$, $y'\to w'$); Group I Rows II and V, Eqs. (5.1), (5.2): $-\frac{\alpha_sN_c}{2\pi}L_\vee\,\mathrm{LO}[\hat{E}_X+\hat{E}_{X'}]$. The sum is

$$\frac{\alpha_sN_c}{2\pi}\,L_\vee\,\mathrm{LO}\Big[\log\frac{(w-w')^4}{(x-w)^2(x'-w')^2}\Big] \qquad \text{(pole and } \mu \text{ cancel, } \texttt{sympy}\text{: 0)},$$

part of $-\mathcal{J}L_\vee$.

**Result.**

$$\boxed{\left.\frac{d^3N}{d^3k}\right|_{\rm DGLAP} = -\frac{\alpha_s}{2\pi}\int_0^1 \frac{d\xi}{\xi^2}\,P^{(r)}_{gg,+}(\xi)\,{\rm LO}_\xi\big[\ell_\Delta(\mu_F)\big]\Big|_{\rm reg}, \qquad \ell_\Delta = \log\frac{(y'-y)^2\mu_F^2}{c_0^2}} \tag{6.3}$$

- "reg": the $\frac{1-\xi}{\xi}$ term at $\xi \to 0$ (the file's lower limit $\frac{k^+}{\vee}$) gives exactly $\mathcal{N}L_\vee\,P_{\rm II.IV}$ with $P_{\rm II.IV} = 2N_c\int_{x,x',z} K(w,w',z)K(x,x',z)\mathcal{O}(z,z)$ (the BFKL piece of Row IV). This uses $\int d^2\Delta\, e^{-i\mathbf{q}\cdot\Delta}(-\ell_\Delta)G(\Delta) = \frac{4\pi}{q^2}G(0)+\dots$. It is part of $\mathcal{B}L_\vee$; "reg" means it is subtracted.
- With $d_\perp = 2$ instead of $2-\epsilon$: $\ell_\Delta \to \ell_\Delta + 1$ (scheme).
- In AABKL's scheme (position-space cutoff $\theta(z^2\mu^2-1)$) the same term has a different constant: $f_{\mu^2} = f^{\overline{MS}}_{1.26\mu^2}$ (their App. C).

# 7 The large logarithm $L_\vee = \log\frac{\vee}{k^+}$

**Where it is.** Besides the explicit $\log\frac{\vee}{k^+}$, $\log\frac{\vee}{k^+\pm\Lambda}$, $\log\frac{k^+-\Lambda}{\vee}$:

1. in the logarithms of Sec. 4 (Group I Rows III, IV; Group II Row I; Group III Rows I, II);
2. in the plus-prescribed integrals up to $\vee - k^+$ (Group III Rows III, V, VI). In the variables of the file ($y'$, after the $w'$ integration) the large-$p^+$ region (region P) is *not* at large $p^+$ for fixed $y'$: pointwise in $y'$ the integrand only oscillates, $\propto e^{-ik\cdot(y'-z)p^+/k^+}$. It sits at $y'\to z$, where $\mathcal{K}(y'-z)$ makes the $y'$ integral behave as $1/p^+$. In the variables $w' = y' + \frac{p^+}{k^+}(y'-z)$, i.e. before the $w'$ integration ($y' = \frac{k^+w'+p^+z}{P}$, $d^2y' = (\frac{k^+}{P})^2 d^2w'$, phase $e^{-ik\cdot(w'-w)}$), the integrand $f(p^+)$ behaves as $\mathcal{T}/p^+$ for $p^+\to 0$ and $\mathcal{P}/p^+$ for $p^+\to\infty$. Then, for any row,

$$\begin{aligned}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}f(p^+) &= \frac{1}{2\pi}\Big[\mathcal{T}\log\frac{\vee}{\Lambda} + (\mathcal{P}-\mathcal{T})L_\vee\Big] \\ &\quad + \int_0^\infty\frac{dp^+}{2\pi}\Big[f(p^+) - \theta(k^+-p^+)\frac{\mathcal{T}}{p^+} - \theta(p^+-k^+)\frac{\mathcal{P}}{p^+}\Big] + O\Big(\frac{\Lambda}{k^+},\frac{k^+}{\vee}\Big);\end{aligned} \tag{7.1}$$

3. in Group II Row IV: Eq. (6.1) (explicit) and the $\xi\to 0$ end of Eq. (6.3).

**Coefficients.** For every row the coefficient of $L_\vee$ is (region-P limit − region-T limit)/$2\pi$ of its $p^+$ integrand, i.e. the $B = P - A$ of `LogVeeKplus_BFKL.pdf`. Checked here directly for the closed forms of Sec. 8 (Group I Row IV, Group II Row I parts 1, 2, Group III Rows I, II: rel. $\le 1\times10^{-6}$) and for Group III Rows III, V, VI in the variables $w'$ (Sec. 8, rel. $\le 2\times10^{-6}$). The sum over all rows, after symmetrizing the three-$\rho$ terms, is

$$\boxed{\left.\frac{d^3N}{d^3k}\right|_{L_\vee} = \mathcal{N}\,L_\vee\,(\mathcal{B}-\mathcal{J})} \tag{7.2}$$

(`LogVeeKplus_BFKL.pdf`: exact, pointwise in $z$, 18 kernel structures; the three-$\rho$ $L_\vee$ terms cancel).

# 8 The remaining finite terms, simplified

Below, each row is written as $C\,L_\vee + R$ with $\Lambda\to 0$, $\vee\to\infty$; $CL_\vee$ is part of Eq. (7.2), and $R$ is the finite rest. The common factor of every row is its prefactor, $\frac{1}{(2\pi)^3}\int e^{-ik\cdot(w'-w)}$ and its colour factor, as in the file. All $R$ were checked against the file at $\vee/k^+ = 10^6, 10^9$, $\Lambda/k^+ = 10^{-7}, 10^{-10}$ (rel. $3\times10^{-6}$ / $3\times10^{-9}$, falling like $k^+/\vee$).

## 8.1 Group I

**Row II.** Part 1: Secs. 5, 6. Part 2a: pure $L_\vee$ (three-$\rho$; cancels after symmetrization). Part 2b:

$$\begin{aligned}\frac{d^3N_{2b}}{d^3k} = &-\frac{1}{(2\pi)^3}\frac{g^4}{64\pi^6}\frac{1}{k^+}\int_{w,w'} e^{-ik\cdot(w'-w)}\int d^2p\,d^2k\int_{x,y,x'} e^{ik\cdot(w-y)}e^{ip\cdot(y-x)}\frac{(x'-w')^i}{(x'-w')^2\mathbf{p}^2\mathbf{k}^2}R^i_\infty \\ &\times\mathcal{C}^a_L\Big[f^{hbc}U^{ah}(w)\{\rho^b(x),\rho^c(y)\} - f^{abc}U^{bd}(x)U^{ce}(y)\{\rho^d(x),\rho^e(y)\}\Big]\end{aligned}$$

**Row III.** Colour $\mathcal{C}^{\rm III} = f^{abd}\mathcal{C}^a_L[U^{de}(x)U^{bc}(z)\rho^e(x)\rho^c(y) - U^{de}(x)U^{bc}(y)\rho^e(x)\rho^c(y)]$, $p_8 = \frac{ig^4}{8\pi^5 k^+}$, $p_{16} = \frac{ig^4}{16\pi^5 k^+}$:

$$
\begin{aligned}
C_{\rm III} &= p_8\Big[-(X'\cdot X_w)(X_z\cdot Y) + (X'\cdot X_w)(Y\cdot Z_w) - \Sigma_6 + T_3\Big],\\
R_{\rm III} &= p_{16}\,(X'\cdot Y)\frac{A+B}{(A-B)^2}\log\frac{A}{B} - p_8(X'\cdot X_w)(X_z\cdot Y)\log\frac{A}{B} - p_8\,\Sigma_6\log\Big|\frac{A}{A-B}\Big|\\
&\quad + p_8\log\Big|\frac{B-A}{B}\Big|\Big[-\frac{(X'\cdot Y)\,Z_w\cdot(x-w)\,A}{(B-A)^2} + \frac{(X'\cdot Y)\,Z_w\cdot(x-z)\,B}{(B-A)^2}\\
&\qquad - \frac{(X'\cdot Z_w)\,Y\cdot(x-w)}{B-A} + \frac{(X'\cdot Z_w)(Y\cdot X_z)B}{B-A}\\
&\qquad + \frac{(Y\cdot Z_w)(X'\cdot X_w)A}{B-A} - \frac{(Y\cdot Z_w)\,X'\cdot(x-z)}{B-A}\Big]\\
&\quad + p_8\,\mathcal{A}_0\Big[(X'\cdot Z_w)\,Y\cdot(x-w) + (Y\cdot Z_w)\,X'\cdot(x-z)\\
&\qquad - \frac{(x-w)\cdot(x-z)}{A}(X'\cdot Z_w)\,Y\cdot(x-z) - \frac{(x-w)\cdot(x-z)}{B}(Y\cdot Z_w)\,X'\cdot(x-w)\Big]\\
&\quad + p_{16}\log\frac{(z-w)^2}{B}\Big[-\frac{(X'\cdot Z_w)\,Y\cdot(x-z)}{A} + \frac{(Y\cdot Z_w)\,X'\cdot(x-w)}{B} - \frac{X'\cdot Y}{(z-w)^2}\Big],
\end{aligned}
$$

with

$$
\begin{aligned}
\Sigma_6 &= \frac{(X'\cdot Z_w)\,Y\cdot(x-w)}{B-A} - \frac{(X'\cdot X_w)(Y\cdot Z_w)A}{B-A} + \frac{(X'\cdot Y)\,Z_w\cdot(x-w)\,A}{(B-A)^2}\\
&\quad - \frac{(X'\cdot Z_w)(Y\cdot X_z)B}{B-A} + \frac{X'\cdot(x-z)\,(Y\cdot Z_w)}{B-A} - \frac{(X'\cdot Y)\,Z_w\cdot(x-z)\,B}{(B-A)^2},\\
T_3 &= \frac{X'\cdot Y}{(z-w)^2} + \frac{(X'\cdot Y)\,Z_w\cdot(x-w)\,A}{(B-A)^2} - \frac{(X'\cdot Y)\,Z_w\cdot(x-z)\,B}{(B-A)^2}.
\end{aligned}
$$

The $\frac{1}{\Lambda}$ terms cancel exactly (E3), and the $\Lambda$ of the arctangents drops.

**Row IV.**

$$
C_{\rm IV} = \frac{ig^4}{8\pi^5 k^+}\Big[X'^i Y^j \mathbb{T}^{ij} - \frac{X'\cdot Y}{2(z-w)^2}\Big], \qquad R_{\rm IV} = \frac{ig^4}{8\pi^5 k^+}\,X'^i Y^j \mathbb{T}^{ij}\log\frac{A}{B}.
$$

**Row V.** Part 1 and parts 2a, 2b finite: unchanged, with $\Lambda \to 0$ in the limits and $\log\frac{k^+-\Lambda}{\vee} \to -L_\vee$. These $L_\vee$ terms are minus the region-T limits ($B = -A$: Row V has no region P). Part 2b UV: Sec. 5.

## 8.2 Group II

**Row I, part 1 (three $\rho$).** Block A:

$$
C = -\frac{ig^4}{16\pi^5 k^+}\Big[X'^i Y^j \mathbb{T}^{ij} - \frac{X'\cdot Y}{2(z-w)^2}\Big], \qquad R = -\frac{ig^4}{16\pi^5 k^+}X'^i Y^j \mathbb{T}^{ij}\log\frac{A}{B};
$$

block B: the same with $w \leftrightarrow w'$ in the kernel and $X' \to \frac{x'-w}{(x'-w)^2}$, and an overall + sign.

**Row I, part 2 (2a–2f, two $\rho$).** Common factor $\frac{1}{(2\pi)^3}\frac{g^4}{8\pi^5}\frac{1}{k^+}\int e^{-ik\cdot(w'-w)}\,\mathcal{C}_{58}$, with $\mathcal{C}_{58}$ the colour of the file and $[\dots]_{2a}$–$[\dots]_{2e}$ the brackets of the file:

$$
\begin{aligned}
&C_{2a} = C_{2d} = C_{2e} = C_{2f} = 0, \qquad C_{2b} = -\frac{[\dots]_{2b}}{BB'}, \qquad C_{2c} = \frac{[\dots]_{2c}}{AA'},\\
&R_{2a} = \frac{[\dots]_{2a}}{D}\log\frac{AB'}{A'B}, \qquad R_{2b} = [\dots]_{2b}\Big(-\frac{A\log\frac{A}{B}}{BD} + \frac{A'\log\frac{A'}{B'}}{B'D}\Big), \qquad R_{2c} = \frac{[\dots]_{2c}}{D}\Big(\frac{B'}{A'}\log\frac{A'}{B'} - \frac{B}{A}\log\frac{A}{B}\Big),\\
&R_{2d} = [\dots]_{2d}\Big(\frac{A\log\frac{A}{B}}{(A-B)D} + \frac{A'\log\frac{A'}{B'}}{(B'-A')D}\Big), \qquad R_{2e} = -[\dots]_{2e}\Big(\frac{B\log\frac{A}{B}}{(A-B)D} + \frac{B'\log\frac{A'}{B'}}{(B'-A')D}\Big),\\
&R_{2f} = \frac{d_\perp}{4}\,\frac{1}{(z-w)^2(z-w')^2}\Big[-\frac{AB(A'-B')}{(A-B)D}\log\frac{A}{B} + \frac{A'B'(A-B)}{(A'-B')D}\log\frac{A'}{B'} - 1\Big].
\end{aligned}
$$

(`sympy`: the $L_\vee$ coefficients of 2d, 2e, 2f are 0; $\frac{d_\perp}{4}$ because 2f has $\frac{g^4 d_\perp}{32\pi^5}$.) The $-1$ in $R_{2f}$ is $k^+(\frac{1}{\vee} - \frac{1}{k^+ + \Lambda})$.

**Row IV.** Lines 1–2: Sec. 6. Line 3 (`RowIV_plus.pdf`: no logarithm at either end):

$$\frac{d^3N_{\rm lin}}{d^3k} = \frac{\alpha_s}{2\pi}\int_0^1 \frac{d\xi}{\xi^2}\, 2N_c\,\xi(1-\xi)\,\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2 k^+}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\Big[X\cdot X' - 2(X\cdot\hat\Delta)(X'\cdot\hat\Delta)\Big]\mathcal{O}, \tag{8.1}$$

$X = \frac{x-y}{(x-y)^2}$, $X' = \frac{x'-y'}{(x'-y')^2}$, $\hat\Delta = \frac{y'-y}{|y'-y|}$. The tensor is traceless: this is the linear polarization of the parent gluon.

## 8.3 Group III

**Rows I, II.** Kernel $C = -C_{\rm IV}$, $R = -R_{\rm IV}$, with the colour of the row.

**Row III part 1, Rows V, VI, and Row III part 2.** Use the variables $w'$ (Eq. (7.1)): $y' = \frac{k^+w'+p^+z}{P}$, $d^2y' = (\frac{k^+}{P})^2 d^2w'$, $e^{-ik\cdot(y'-w)}e^{-ik\cdot(y'-z)\frac{p^+}{k^+}} = e^{-ik\cdot(w'-w)}$. Each row is

$$\frac{1}{(2\pi)^3}\int_{w,w',x',z,y,x} e^{-ik\cdot(w'-w)}\left\{\frac{1}{2\pi}\Big[(\mathcal{P}_C-\mathcal{T}_C)L_\vee\Big] + \int_0^\infty\frac{dp^+}{2\pi}\Big[f_C(p^+) - \theta(k^+-p^+)\frac{\mathcal{T}_C}{p^+} - \theta(p^+-k^+)\frac{\mathcal{P}_C}{p^+}\Big]\right\} \tag{8.2}$$

(plus $\frac{1}{2\pi}\mathcal{T}_C\log\frac{\vee}{\Lambda}$, which is in your evolution terms). Here $f_C(p^+) = (\frac{k^+}{P})^2\times$[the $p^+$ integrand of the file, without its $\frac{1}{2\pi}$, with the plus prescription removed] at $y' = \frac{k^+w'+p^+z}{P}$, and, with $c_{\rm III(1)} = \frac{ig^4}{8\pi^4}$, $c_{\rm V} = c_{\rm VI} = \frac{ig^4}{4\pi^4}$ (the $f$'s are in the colour factors):

$$\mathcal{T}_C = -\frac{c_C}{k^+}\,\big[\mathcal{K}(w'-z)\cdot\mathcal{K}(x-z)\big]\big[\mathcal{K}(y-w)\cdot\mathcal{K}(x'-w')\big]\ \text{colour}\big|_{y'=w'},$$
$$\mathcal{P}_C = -\frac{c_C}{k^+}\,\big[\mathcal{K}(w'-z)\cdot\mathcal{K}(y-w)\big]\big[\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\big]\ \text{colour}\big|_{y'=z};$$

and for Row III part 2 ($c = \frac{g^4}{4\pi^4}$):

$$\mathcal{T} = -\frac{c}{k^+B}\Big[\frac{\mathcal{K}(x'-w')\cdot(x-w)\,\mathcal{K}(w'-z)\cdot(z-w)}{(z-w)^2} + \frac{\mathcal{K}(x'-w')\cdot(x-w)\,\mathcal{K}(w'-z)\cdot(x-z)}{2A}\Big]\text{colour}\big|_{y'=w'},$$
$$\mathcal{P} = -\frac{c}{k^+A}\Big[\frac{\mathcal{K}(x'-z)\cdot(x-z)\,\mathcal{K}(w'-z)\cdot(z-w)}{(z-w)^2} - \frac{\mathcal{K}(x'-z)\cdot(x-z)\,\mathcal{K}(w'-z)\cdot(x-w)}{2B}\Big]\text{colour}\big|_{y'=z}.$$

- $\mathcal{T}$ is the coefficient of your $\log(1-\frac{k^+}{\vee})$ term (times $-\frac{1}{2\pi}$); $\mathcal{P}$ comes from the $\frac{P}{k^+}\delta_{im}\delta_{jk'}$ part of the vertex (Row III part 2: from the $\frac{p^+}{k^{+2}}$ bracket), where the parent sits at $y' = z$. This is the region-P term of `LogVeeKplus_BFKL.pdf`: the measured gluon $w'$ is emitted softly by the parent at $z$.
- Limits of $p^+f_C$ vs. $\mathcal{T}$, $\mathcal{P}$: rel. $\le 2\times10^{-6}$. Eq. (8.2) vs. the exact $\int_\Lambda^{\vee-k^+}$: rel. $4\times10^{-2}$ / $4\times10^{-5}$ at $\vee/k^+ = 10^3$ / $10^5$.
- **Caution.** Done pointwise in $y'$, the limit $\vee\to\infty$ of the file gives closed forms ($\int_0^\infty dp\,\frac{e^{-iap}}{p+k} = e^{iak}E_1(iak)$, $\int_0^\infty dp\,\frac{e^{-iap}-\theta(k-p)}{p} = -\gamma_E - \log(iak)$, $\int_0^\infty dp\,e^{-iap} = \frac{1}{i(a-i0)}$, $a = \frac{k\cdot(y'-z)}{k^+}$). They are correct pointwise, but the last one, times $\mathcal{K}(y'-z)$, is not integrable at $y'\to z$: it has lost $\mathcal{P}\,L_\vee$. Use Eq. (8.2).

# 9 Three-$\rho$ terms

**Symmetrized parts.** With commuting charges (the symmetrized part $S(\rho\rho\rho)$):

1. **Group I Row IV + Group III Row I = 0** exactly, at the level of the $p^+$ integrand, for every $p^+$ and every configuration, also for a general $U$ (max. rel. $6\times10^{-15}$). So their finite parts cancel as well.
2. **Group II Row I part 1 (block A) + Group III Row II** = (kernel of Group III Rows I, II)$\times\big(-\mathcal{C}_L^a\rho^b(y)\mathcal{R}^{ab}(w,z,x)\big)$ (rel. $4\times10^{-16}$). Simplified:

$$\frac{1}{(2\pi)^3}\int e^{-ik\cdot(w'-w)}\int_{x,y,z,x'}\frac{ig^4}{8\pi^5k^+}\Big[X'^iY^j\mathbb{T}^{ij}\Big(L_\vee + \log\frac{A}{B}\Big) - \frac{X'\cdot Y}{2(z-w)^2}L_\vee\Big]\mathcal{C}_L^a\rho^b(y)\,\mathcal{R}^{ab}(w,z,x) \tag{9.1}$$

   Block B and the c.c. of Row II are its mirror image.
3. **The rest does not cancel.** Example: the network $f^{abd}\mathcal{C}_L^a(U(z)\rho(y))^b(U(x)\rho(x))^d$ appears in Group I Row III and in Eq. (9.1). Its kernels differ by $O(1)$ (0.6–8.6 relative): the virtual row has the energy denominator $p^+A - k^+B$, the real rows $p^+A + k^+B$. The two agree only in the soft limits, so the log $\frac{\vee}{\Lambda}$ and $L_\vee$ parts cancel (earlier notes), the finite parts do not.

So the symmetrized three-$\rho$ finite terms are a genuine $O(g^4\rho^3)$ contribution: Group I Row II 2b, Row III, Row V part 1, Eq. (9.1) and its mirror, Group III Row III part 1, Rows V, VI (Group I Row IV and Group III Row I cancel). This is the classical first saturation correction of the projectile (Chirilli, Kovchegov, Wertepny, arXiv:1508.07962; Kovchegov, Skokov, arXiv:1802.08166). It is odd in $\rho$, so it vanishes for a projectile whose charge distribution is Gaussian (MV), $\langle S(\rho\rho\rho)\rangle = 0$, but not for, e.g., a valence quark ($\propto d^{abc}$). See Q2.

**Commutator parts.** With $[\rho^a(u), \rho^b(v)] = if^{abc}\rho^c(u)\delta^{(2)}(u-v)$ and the identity of `ThreeRho_to_TwoRho.pdf`,

$$ABC = S(ABC) + \tfrac{1}{4}\big(\{[A,B],C\} + \{[A,C],B\} + \{[B,C],A\}\big) + (\text{one-}\rho),$$

a three-$\rho$ term $\int \mathcal{K}(u_1,u_2,u_3)\, C^{abc}\rho^a(u_1)\rho^b(u_2)\rho^c(u_3)$ (in the operator order of the file, c.c. terms reversed, $A\{B,C\}$ as $\frac{1}{2}$ of the sum) has the two-$\rho$ part

$$\frac{i}{4}\int \Big[\mathcal{K}(u,u,v)\, f^{abe}C^{abc}\{\rho^e(u),\rho^c(v)\} + \mathcal{K}(u,v,u)\, f^{ace}C^{abc}\{\rho^e(u),\rho^b(v)\} + \mathcal{K}(v,u,u)\, f^{bce}C^{abc}\{\rho^a(v),\rho^e(u)\}\Big]. \quad (9.2)$$

The colour factors $C^{abc}$ are those of the three-$\rho$ rows, so the colour algebra is that of the three-$\rho$ $\to$two-$\rho$ list of `Evolution_terms.tex` (Sec. 2), which you have and which was checked (`Final_evolution_check.pdf`). Hence:

> Two-$\rho$ finite terms from three $\rho$'s = that list, row by row, with the kernel $\log\frac{\vee}{\Lambda}\times\mathcal{K}^{\rm evol}_{\rm row}$ replaced by $R_{\rm row}$ of Sec. 8, at the same coincident points.

The $L_\vee$ parts ($C_{\rm row}$) of these commutator terms are already in Eq. (7.2).

# 10 Final result

For hadron production ($\overline{MS}$, $d_\perp = 2-\epsilon$, $n_f = 0$, rule A for the rapidity logarithms):

$$\begin{aligned}
\frac{d^3N_h}{d^3k} &= \int\frac{d\zeta}{\zeta^2}\,D_{h/g}(\zeta,\mu_F)\,\frac{d^3N_g}{d^3k}\Big(\frac{k^+}{\zeta},\frac{\mathbf{k}}{\zeta}\Big),\\
\frac{d^3N_g}{d^3k} &= \underbrace{\mathrm{LO}\big|_{\alpha_s(\mu_F)}}_{\text{LO}} + \underbrace{\mathcal{N}\log\frac{\vee}{\Lambda}\,\mathcal{J}}_{\text{evolution (JIMWLK)}} + \underbrace{\mathcal{N}\log\frac{\vee}{k^+}\,(\mathcal{B}-\mathcal{J})}_{\text{Eq. (7.2)}}\\
&\quad + \underbrace{\frac{\alpha_s}{2\pi}N_c\Big(\frac{67}{9}-\frac{\pi^2}{3}\Big)\mathrm{LO}}_{\text{running coupling, Eq. (5.3)}} \underbrace{-\frac{\alpha_s}{2\pi}\int_0^1\frac{d\xi}{\xi^2}P^{(r)}_{gg,+}(\xi)\,\mathrm{LO}_\xi\big[\ell_\Delta(\mu_F)\big]\Big|_{\rm reg}}_{\text{DGLAP, Eq. (6.3)}}\\
&\quad + \underbrace{\frac{d^3N_{\rm lin}}{d^3k}}_{\text{Eq. (8.1)}} + \underbrace{\sum_{\text{two-}\rho\text{ rows}} R}_{\text{G1 V(2), G2 I(2), G3 III(2)}} + \underbrace{\sum_{\text{rows}}[\text{Eq. (9.2)}]}_{\text{two-}\rho\text{ from three }\rho} + \underbrace{\sum_{\text{three-}\rho\text{ rows}} S(R)}_{\text{Sec. 9}}\\
&\quad + O(\alpha_s^3) + O\Big(\frac{\Lambda}{k^+},\frac{k^+}{\vee}\Big).
\end{aligned} \quad (10.1)$$

- LO is $O(\alpha_s)$; every other term is $O(\alpha_s^2)$.
- No $\Lambda$ remains except in $\log\frac{\vee}{\Lambda}$. No pole and no $\mu$ remain except $\mu_F$ (of $D$ and $\alpha_s$).
- The explicit two-$\rho$ rests $R$ are in Sec. 8: Group I Row V part 2 (2a, 2b finite), Group II Row I part 2 ($R_{2a}$–$R_{2f}$) and Group III Row III part 2.
- With rule C instead of A, the second line reads $\mathcal{N}[\frac{1}{2}\log\frac{\vee}{\Lambda}(\mathcal{J}+\mathcal{B}) + \frac{1}{2}\log\frac{k^{+2}}{\vee\Lambda}(\mathcal{J}-\mathcal{B})]$. At mid rapidity the second term vanishes.

# 11 Comparison with AABKL (arXiv:1411.2869)

**Setting.** AABKL: forward production in the hybrid formalism. The projectile is a collinear parton $x_p f(x_p)$ with zero transverse momentum, LO is $O(\alpha_s^0)$ ($s_A[y,\bar{y}]$), NLO is one splitting. Yours: mid rapidity. The projectile charges $\rho$ emit the gluon (LO is $O(\alpha_s)$), and NLO is the splitting of that gluon, its virtual corrections, and the corrections involving the charges. The natural comparison is their gluon channel, $g\to g$: the paper's Eqs. (4.9), (4.10), (4.22), (6.8)–(6.19). The incoming gluon there corresponds to the gluon emitted by $\rho$ here, the parent of momentum $P = p^+ + k^+$.

| AABKL | your result | comment |
|---|---|---|
| $F_{(gg)} \propto$ $(1-\xi)\delta_{im}\delta_{jk}+\xi\delta_{jm}\delta_{ik}-\xi(1-\xi)\delta_{km}\delta_{ij}$, Eq. (6.9) | vertex of $\mathfrak{C}_1$: $\delta_{km}\delta_{ij} - \frac{P}{p^+}\delta_{jm}\delta_{ik} - \frac{P}{k^+}\delta_{im}\delta_{kj}$ | identical after multiplying by $-\xi(1-\xi)$ ($\xi_{\text{AABKL}} = p^+/P$) |
| $P_{gg}$: $\frac{1-\xi}{[\xi]_+} + \frac{\xi}{1-\xi} + \xi(1-\xi)$ | $P^{(r)}_{gg,+}$, Eq. (6.3) | same (their $\xi$ is the unobserved fraction) |
| real, splitting before the target: $2C_A\, A\cdot A\, s_A[y,\bar{y}]$ | G2 Row I part 2 ($\mathfrak{B}^\dagger\mathfrak{B}$, first term of $\mathcal{C}_{58}$: $U(w')U(w)$) | theirs needs the PDF subtraction $C_{\mu^2}$; yours is finite: the parent is emitted by $\rho$ at distance $\lvert x-w\rvert$, the energy denominators $p^+A + k^+B$ cut off large $z$ |
| real, splitting after the target: $(1-\xi)^2 s_A[(1-\xi)y,(1-\xi)\bar{y}]$ | G2 Row IV ($\mathfrak{C}^\dagger\mathfrak{C}$), parent at $y = \xi w + (1-\xi)z$ | both give the FF counterterm; your finite rest is Eq. (6.3) |
| interference: $\frac{N_c^3}{N_c^2-1}\{sss\} - \frac{1}{N_c^2}\bar{X}$, Eq. (6.15) | G3 Row III part 2 ($\mathfrak{C}^\dagger\mathfrak{B}$) | same topology |
| — | G2 Row I part 2, the other terms of $\mathcal{C}_{58}$ (charge rotated, $U(x)$) | the parent emitted *after* the target by the rotated charge: no analogue (their parent always exists before the target) |
| virtual dressed gluon, Eq. (6.17): $-N_c s_A + \frac{N_c^3}{2(N_c^2-1)}(sss)$ | G1 Row V part 2a: $-ffU(x)U(z) + N_cU(x)$ | same structure ($D(z_1,z_2) - N_cS$ of KLSZ) |
| — | G1 Row II part 1, Row V 2b UV ($\beta_0$, CMW) | no coupling renormalization in AABKL (LO is $\alpha_s^0$); yours: $\beta_0$ cancels, Eq. (5.3) |
| unpolarized parent: $A^iA^i$ only | Eq. (8.1), $[X\cdot X' - 2(X\cdot\hat{\Delta})(X'\cdot\hat{\Delta})]$ | your parent is linearly polarized along the WW field of $\rho$; absent in AABKL |
| Ioffe time: $\frac{2\xi(1-\xi)x_pP^+}{l_\perp^2} > \tau$, fields $A_\xi$ with $J_0$ | cutoffs $\Lambda < p^+ < \vee$, exact energy denominators | your logs $\log\frac{(\vee-k^+)A+k^+B}{\Lambda A+k^+B} \to L_\vee + \log\frac{A}{B}$ play the role of their $\xi_{\min} = \max\{\frac{l^2}{x_ps_0}, \frac{m^2}{x_ps_0}\}$ (Eqs. (3.11)–(3.13)) |
| last term of (4.22): $\int\frac{d\xi}{\xi}[A_\xi(y-z) - A_\xi(\bar{y}-z)]^2[\dots]$ ("not extra evolution") | $\mathcal{N}\log\frac{\vee}{\Lambda}\mathcal{J} + \mathcal{N}L_\vee(\mathcal{B}-\mathcal{J})$ | their soft region $\xi\to 0$ = your region T (target, JIMWLK/BK); your region P (BFKL) has no analogue: their parent momentum is fixed by the PDF |
| scheme: $C_{\mu^2}$, $\theta(z^2\mu^2-1)$; $f_{\mu^2} = f^{\overline{MS}}_{1.26\mu^2}$ | $\overline{MS}$, $d_\perp = 2-\epsilon$ | constants differ by a rescaling of $\mu$ |
| — | three-$\rho$ terms, Sec. 9 | no analogue (single parton) |

**In words.**

- Where the parent gluon can be treated as an on-shell incoming parton, the emission by $\rho$ factorizes from the splitting, and your Groups II and III reduce to the AABKL $g\to g$ formulas. This is the region where the transverse size of the emission, $\lvert x-w\rvert$, is large compared with $\lvert w-z\rvert$ (energy denominator of the parent $\ll$ that of the pair).
- Outside it, your result contains terms AABKL do not have: the emission after the target by the rotated charge, the interplay of $p^+A$ and $k^+B$ in the energy denominators, the linear polarization, and the three-$\rho$ terms.
- So "resembles" is right in structure: the same vertex, splitting function, colour topologies, virtual term and FF subtraction. It is not right term by term.
- Two genuine differences matter for the final formula:
  - AABKL have no BFKL logarithm (projectile evolution sits in the PDF), while you have $\mathcal{N}L_\vee\mathcal{B}$;
  - AABKL's evolution interval depends on $l_\perp$ through the Ioffe time; yours is fixed ($\log\frac{\vee}{\Lambda}$) and the $l_\perp$ dependence sits in the finite logs $\log\frac{A}{B}$.

# 12 Questions

**Q1 Rule A leaves $\mathcal{N}L_\vee(\mathcal{B}-\mathcal{J})$ in the "finite" terms.** It is not finite as $\vee\to\infty$; at mid rapidity it is $\frac{1}{2}\log\frac{\vee}{\Lambda}(\mathcal{B}-\mathcal{J})$. Do your collaborators accept that (and call it finite), or should $\mathcal{N}L_\vee\mathcal{B}$ be counted as projectile (BFKL) evolution (rule B/C)?

**Q2 The symmetrized three-$\rho$ finite terms do not cancel** (Sec. 9). Keep them as the $O(g^4\rho^3)$ (first saturation) correction, or drop them? Dropping is justified for a Gaussian projectile, or if you count powers of $g\rho$ (dilute projectile). If you expected them to cancel, a contribution is missing; the soft limits do cancel.

**Q3** **The two-$\rho$ terms from the three-$\rho$ finite terms** (Eq. (9.2)): do you want them written out row by row (as long as the evolution list), or is the rule enough?

**Q4** **Group I Row I′:** the diagonal window $|p^+ - k^+| < \Lambda$ of $\bar{\mathbb{B}}_3^\dagger$ (the measured gluon emits and reabsorbs a soft gluon after the target). Your file says Row I: NA. Is its finite part included anywhere? It was an open item in the draft.

**Q5** **Upper limits:** virtual rows run to $\vee$, real rows to $\vee - k^+$. This leaves $O(k^+/\vee)$ rests (F6, and $-N_c \log(1 - \frac{k^+}{\vee})$ in the $\delta(1-\xi)$ tally). Drop them, or make the limits consistent?

**Q6** **Scheme:** $d_\perp = 2 - \epsilon$ (your $W$) or $d_\perp = 2$? This decides whether $\ell_\Delta$ or $\ell_\Delta + 1$ appears in Eq. (6.3); it has to match the FF set used.

**Q7** **Observable:** gluons or hadrons? The gluon spectrum alone is not collinear safe; Eq. (10.1) is for hadrons.

**Q8** **Coupling in LO:** $\alpha_s(\mu_F)$, or KLSZ's $g(c_0^2/X^2)g(c_0^2/X'^2)$ with the compensating $-\frac{\alpha_s\beta_0}{8\pi}(\ell_X + \ell_{X'})$ (Sec. 5)?

**Q9** **Ioffe time:** do your collaborators want AABKL's Ioffe-time restriction imposed (cutoff in $\xi$ and $l_\perp$) instead of the $p^+$ cutoffs? It would change the finite terms at $O(1)$ (their Sec. III.C).

**Q10** Confirm the measure of Group III Row III part 2 (F4).

# 13 Checks

Scripts in `nlo_gluon_paper/finite/`: `ft.py` (the file transcribed), `verify.py` (Table 1), `simp.py` (Sec. 8), `crows_w.py` (Group III Rows III, V, VI in $w'$ variables, → `crows_w.json`), `checks.py` (everything else) → `checks.json`.

| check | result |
|---|---|
| file vs. exact $p^+$ integrals, closed-form rows | spread $\leq 7 \times 10^{-13}$ |
| file vs. exact, plus-prescription rows (lower limit 0) | spread $\leq 1 \times 10^{-4} = O(\Lambda/k^+)$ |
| simplified forms vs. file, $\vee/k^+ = 10^6$ / $10^9$ | $3 \times 10^{-6}$ / $3 \times 10^{-9}$ |
| $L_\vee$ coefficient = (region P − region T)/$2\pi$ | rel. $\leq 1 \times 10^{-6}$ |
| Group III Rows III, V, VI: $\mathcal{T}$, $\mathcal{P}$ vs. limits of $p^+ f$ ($w'$ variables) | $\leq 2 \times 10^{-6}$ |
| Group III Rows III, V, VI: Eq. (8.2) vs. exact integral, $\vee/k^+ = 10^3$ / $10^5$ | $4 \times 10^{-2}$ / $4 \times 10^{-5}$ |
| Group I Row IV + Group III Row I (integrand, symmetric and general $U$) | $6 \times 10^{-15}$ |
| Group II Row I part 1 (A) + Group III Row II = Eq. (9.1) | $4 \times 10^{-16}$ |
| $U(z)$ network: Group I Row III + Eq. (9.1) (does not cancel) | 0.6–8.6 |
| $L_\vee$ coefficients of 2b, 2c (vs. $C_{2b}$, $C_{2c}$), 2d, 2e, 2f (`sympy`) | 0, 0, 0, 0, 0 |
| $\beta_0$ tally: Row II, Row V 2b UV (units $\frac{\alpha_s\beta_0}{4\pi}\hat{E}$); sum | 2, −2; 0 |
| $L_\vee \hat{E}$ tally (Rows II, V, IV), constant | 0; $\frac{\alpha_s}{2\pi} N_c(\frac{67}{9} - \frac{\pi^2}{3})$ |